Ans. (a) : A consumer will be said to be in equilibrium when he is able to meet his needs at a certain level of his income. In case of deviation of equilibrium, there has been a situation of imbalance between the income and needs of the consumer.

68.

Year	Production of Rice and Wheat (In millions of Tonnes)		
	Rice	Wheat	Percentage of Wheat to Rice
1950-51	20.58	6.46	31.4
1960-61	34.58	11.00	31.8
1970-71	42.22	23.83	56.4
1980-81	58.63	36.31	67.7
1990-91	74.29	55.14	74.2
1994-95	81.81	65.77	80.4
1995-96	79.62	62.62	78.6

The above table indicates the performance of India in rice and wheat production form 1950-51 to 1995-96. Which of the following conclusions arrived at from the above table would be valid?

1. **Record production of rice as well as wheat has been in 1994-95.**
2. **The ratio of wheat to rice production seems to have steadily increased over 16 years.**
3. **Wheat has not been popular among the Indian population before 1980.**
4. **India became self-sufficient in rice and wheat only after 1990.**

Select the correct answer using the codes give below:

Codes:

(a) 1 and 2 (b) 1, 2, 3 and 4
(c) 3 and 4 (d) None

Ans. (a) : It is clear from the table that in the year 1994-95, the record production of rice and wheat in comparison to the previous year was 81.81 million tonnes (rice) and 65.77 million tonnes respectively. The ratio of rice production to wheat production has steadily increased over a period of 16 years.

69. The supply-side economics lays greater emphasis on the point of view of:

(a) Producer (b) Global economy
(c) Consumer (d) Middle-man

Ans. (a) : Supply side economics lays greater emphasis on the point of view of the producer whereas the demand side economics lays emphasis interest of the consumer.

PHYSICS

70. Match the names of outstanding Indian scientists given in List-I with area of their specialized work given in List-II and select the correct answer by using the codes given below the lists:

List-I	List-II
A. Dr. Raja Ramanna	1. Plant Chemistry
B. Dr. M.S. Swaminathan	2. Nuclear Physics
C. Prof. U.R. Rao	3. Thermodynamics and astrophysics
D. Prof. Meghnad Saha	4. Space research
	5. Agricultural sciences

Codes:

	A	B	C	D
(a)	3	5	2	1
(b)	2	1	4	3
(c)	2	5	4	3
(d)	3	1	4	2

Ans. (c) : Raja Ramanna was an Indian physicist who is best known for his role in India's nuclear program during its early stages. Having joined the nuclear program in 1964, Ramanna worked under Homi Jahangir Bhabha, and later became the director of this program in 1967.

Mankombu Sambasivan Swaminathan is an Indian geneticist and administrator, known for his role in India's Green Revolution, a program under which high-yield varieties of wheat and rice were developed.

Prof U.R. Rao was an internationally renowned space scientist who had made original contributions to the development of space technology in India and its extensive application to communications and remote sensing of natural resources.

Meghnad Saha FRS was a Indian astrophysicist best known for his development of the Saha ionization equation, used to describe chemical and physical conditions in stars.

71. One Astronomical Unit is the average distance between:

(a) Earth and the Sun
(b) Earth and the Moon
(c) Jupiter and the Sun
(d) Pluto and the Sun

Ans. (a) : We can say that one astronomical unit (AU) represents the mean distance between the Earth and our sun. An AU is approximately 93 million miles (150 million km). It's approximately 8 light-minutes. More exactly, one astronomical unit (AU) = 92,955,807 miles (149,597,871 km).

72. Consider the following statements regarding asteroids:

1. **Asteroids are rocky debris of varying sizes orbiting the Sun.**
2. **Most of the asteroids are small but some have diameter as large as 1000 km.**
3. **The orbit of asteroids lies between the orbits of Jupiter and Saturn.**

Of these statements–

(a) 1, 2 and 3 are correct
(b) 2 and 3 are correct
(c) 1 and 2 are correct
(d) 1 and 3 are correct

Ans. (c) : Asteroids are rocky debris of varying sizes with diameter from a few kilometres to as large as 1000 km. Their orbit is between Mars and Jupiter and not between Jupiter and Saturn.

73. A ball is dropped from the top of a high building with a constant acceleration of 9.8 m/s^2. What will be its velocity after 2 seconds?

(a) 9.8 m/s (b) 19.6 m/s
(c) 29.4 m/s (d) 39.2 m/s

Ans. (b) : The velocity after time t is given by $v = u + gt$
Here, $u = 0$, $t = 2s$, $g = 9.8\ m/s^2$
$v = 0 + (9.8 \times 2) = 19.6$ m/s

CHEMISTRY

74. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A. Blue Vitriol	1. Sodium bicarbonate
B. Epsom salt	2. Sodium hydroxide
C. Baking soda	3. Magnesium sulphate
D. Caustic soda	4. Copper sulphate

Codes:

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	2	1
(c)	3	4	2	1
(d)	4	3	1	2

Ans. (d) : Blue vitriol is blue, crystalline hydrous solution of copper sulphate, $CuSO_4 \cdot 5H_2O$, one of the most important industrial copper salts, used in insecticides, germicides, and hair dyes and in the processing of leather and textiles. Magnesium sulphate is a chemical compound containing magnesium, sulphur and oxygen, with the formula $MgSO_4$. It is often encountered as the heptahydrate epsomite ($MgSO_4 \cdot 7H_2O$), commonly called "Epsom salt".

Sodium bicarbonate or sodium hydrogen carbonate is the chemical compound with the formula $NaHCO_3$. The salt has many related names such as baking soda, bread soda, cooking soda, bicarbonate of soda. Caustic soda or sodium hydroxide is an essential ingredient in an array of industrial applications. In addition, consumers use caustic soda when using cleaners, such as oven and drain cleaners.

75. Some time back, the Government of India, decided to delicense 'white goods' industry. 'White goods' include:

(a) Stainless steel and aluminium utensils
(b) Milk and milk products
(c) Items purchased for conspicuous consumption
(d) Soaps, detergents and other mass consumption goods

Ans. (c) : White goods are large electrical goods used domestically such as air conditioners, refrigerators, cooking range etc., which had a white enamel finish. Despite their availability in varied colours now, they are still called white goods. The term is used where British English is spoken. In the year 1993 the govt. of India had decided to de-licence these goods.

76. Examine the following three statements:

1. **Processed meat a perishable food.**
2. **All perishable foods are packed in sealed tins.**
3. **Sealed tins sometimes do not contain processed meat.**

Which one of the following inferences can be drawn from the above statements?

(a) Sealed tins always contain perishable food
(b) Processed meat is sometimes not packed in sealed tins processed meat is always packed in sealed tins
(c) Processed meat is always packed in sealed tins
(d) Non-perishable foods are never packed in sealed tins

Ans. (c) : We can draw the inference that processed meat is always packed in sealed tins.